## 2.4-2.5 other elasticities

EECDC
CCEDB
DDCAC
ADDBD
AD

**1.** A decrease in the demand for antique lanterns, a normal good, would be caused by which of the following?

(A) An increase in consumers' incomes
(B) An increase in the price of antique lanterns
(C) An increase in the expected price of antique lanterns in the future
(D) An increase in the price of electric lanterns, a substitute good
(E) An increase in the price of lamp oil, a complementary good ✓

**Answer E**

Correct. An increase in the price of a complement in consumption (lamp oil) will decrease demand for antique lanterns.

**2.** If the price elasticity of supply for pickles is $2$ and the price of pickles increases by $10$ percent, then the quantity supplied of pickles will increase by

(A) $0.2\%$
(B) $5\%$
(C) $8\%$
(D) $12\%$
(E) $20\%$ ✓

**Answer E**

Correct. The price elasticity of supply is defined as the percentage change in the quantity supplied of a good divided by the percentage change in the price of the good. Given the value for the elasticity and the percentage change in the price, the resulting percentage change in the quantity supplied can be determined as follows: percentage change in quantity supplied $=$ (elasticity of supply $\times$ the percentage change in price) $= (2 \times 10\%) = 20\%$.

**3.** If a $10$ percent increase in the price of good X results in a $20$ percent decrease in the quantity of good Y demanded, which of the following is true?

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(A) Good X and good Y are complementary goods, and the cross-price elasticity is $-0.5$ .
(B) Good X and good Y are substitute goods, and the income elasticity is $+2$ .
(C) Good X and good Y are complementary goods, and the cross-price elasticity is $-2$ . ✓
(D) Good X and good Y are normal goods, and the income elasticity is $+2$ .
(E) Good X and good Y are substitute goods, and the cross-price elasticity is $-2$ .

**Answer C**

Correct. The cross-price elasticity between good X and Y $(\mathrm{CRXY})$ can be calculated as the ratio of the percentage change in the quantity of good Y to the percentage change in the price of good X. Thus, $\mathrm{CRXY} = (-20\,\% / 10\,\%) = -2$. The cross-price elasticity between good X and good Y is negative implying that X and Y are complementary goods.

**4.** At the current prices of goods X and Y, the quantity demanded of good X is 10 units, and the quantity demanded of good Y is 5 units. The cross-price elasticity of demand between goods X and Y is 0.6. A 10 percent increase in the price of good Y will result in which of the following?

(A) A 0.5 percent decrease in the quantity demanded of good Y.
(B) A 3 percent increase in the quantity demanded of good Y.
(C) A 1 percent increase in the quantity demanded of good X.
(D) A 6 percent increase in the quantity demanded of good X. ✓
(E) A 60 percent increase in the quantity demanded of good X.

**Answer D**

Correct. The cross-price elasticity of demand between goods X and Y is 0.6, which is the ratio of the percentage change in the quantity of good X demanded to the percentage change in the price of good Y. This implies that the 10 percent increase in the price of good Y will result in a 6 percent increase in the quantity demanded of good X. This value is obtained by multiplying the cross-price elasticity by the percentage change in the price of good Y. That is, 0.6 times 10 percent equals 6 percent.

**5.** Assume that a 2 percent increase in the price of bologna causes a 5 percent decrease in the demand for cheese. What is the cross-price elasticity of demand between the two goods, and how are these goods related?

### 2.4-2.5 other elasticities

(A) Cross-price elasticity of demand equals $-0.4$, and these goods are complements.
(B) Cross-price elasticity of demand equals $+0.4$, and these goods are substitutes.
(C) Cross-price elasticity of demand equals $-2.5$, and these goods are complements. ✓
(D) Cross-price elasticity of demand equals $+2.5$, and these goods are substitutes.
(E) Cross-price elasticity of demand equals $-0.4$, and these goods are substitutes.

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**The table below shows the values of different elasticities of demand for good J at the market equilibrium price.**

| | |
|---|---|
| Price elasticity (absolute value) | 0.75 |
| Cross-price elasticity with respect to good X | 0.8 |
| Cross-price elasticity with respect to good Y | 2 |
| Cross-price elasticity with respect to good Z | $-2.5$ |
| Income elasticity | $-0.5$ |

**6.** Which of the following would result in the greatest rightward shift of the demand curve for good J?

(A) A $50\%$ decrease in the price of good J.
(B) A $20\%$ increase in the price of good X.
(C) A $10\%$ increase in the price of good Y. ✓
(D) A $10\%$ increase in the price of good Z.
(E) A $10\%$ decrease in income.

**Answer C**

Correct. Goods J and Y are substitutes in consumption since the cross-price elasticity with respect to good Y is positive. Therefore, a $10\%$ increase in the price of good Y will shift the demand for good J to the right by $20\%(=$ % change in price of good Y $\times$ cross-price elasticity with respect to good Y $= 10\% \times 2)$. This is the largest increase among the answer choices.

**7.** Which of the following is true about good J?

(A) Good J is a normal good.
(B) Good J's demand is elastic.
(C) Good J is an inferior good. ✓
(D) Good J is a complement in consumption to good Y.
(E) Good J is a substitute in consumption to good Z.

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## 2.4-2.5 other elasticities

**Answer C**

Correct. Whether a good is inferior or normal is based on the sign of the income elasticity – with a positive sign indicating that the good is a normal good and a negative sign indicating that the good is an inferior good. The income elasticity of good J is negative $(-0.5)$; therefore, good J is an inferior good.

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**8.** The cross-price elasticity of demand between goods J and K is $-3$. A 20 percent decrease in the price of good K will result in a

(A) 3 percent decrease in the quantity demanded of good K
(B) 15 percent decrease in the quantity demanded of good K
(C) 6 percent increase in the quantity demanded of good J
(D) 12 percent increase in the quantity demanded of good J
(E) 60 percent increase in the quantity demanded of good J ✓

**Answer E**

Correct. The cross-price elasticity of demand between goods J and K is $-3$, which is the percentage change in the quantity demanded of good J divided by the percentage change in the price of good K. This implies that the 20 percent decrease in the price of good K will result in a 60 percent increase in the quantity demanded of good J, which is calculated as $-3$ times $-20$ percent equals 60 percent.

**9.** The cross-price elasticity of demand for Goods X and Y is $-2.8$. Based on this information, what is true about Good X and Good Y?

(A) Good X is a normal good and Good Y is an inferior good.
(B) The goods are inferior.
(C) The goods are substitutes.
(D) The goods are complements. ✓
(E) The goods have upward-sloping demand curves.

**Answer D**

Correct. A negative cross-price elasticity of demand coefficient indicates that Goods X and Y are

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## 2.4-2.5 other elasticities

complements. Since the two goods are consumed together, an increase in the price of one good will cause a decrease in the demand, and therefore the quantity demanded, for the other good.

**10.** Which of the following is true of the cross-price elasticity of demand?

(A) It can indicate if a good is a necessity or a luxury.

(B) It is greater than zero for two goods that are substitutes. ✓

(C) It is close to zero if the two goods are closely related.

(D) It is always negative because demand curves are downward sloping.

(E) It increases as income increases.

**11.** In relation to tacos, the cross-price elasticity of demand for Good X is negative. What would be the result if the price of tacos increases?

(A) There will be a movement up the demand curve for Good X because it is a complement to tacos.

(B) There will be a movement down the demand curve for Good X because it is a complement to tacos.

(C) The demand for Good X will increase because it is a substitute for tacos.

(D) The demand for Good X will decrease because it is a complement to tacos. ✓

(E) The demand for Good X will increase because tacos and Good X are normal goods.

**Answer D**

Correct. When the cross-price elasticity is negative, this means that the two products are complements. Therefore, an increase in the price of tacos will decrease the demand for Good X.

**12.** To determine whether two goods are complements, one would calculate the

(A) price elasticity of demand

(B) price elasticity of supply

(C) income elasticity of demand

(D) cross-price elasticity of demand ✓

(E) input-price elasticity of supply

**13.** The cross-price elasticity of demand between good X and good Z measures the percentage change in the quantity demanded of good X in response to a percentage change in

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(A) the price of good X
(B) income
(C) the price of good Z ✓
(D) the supply of good Z
(E) total expenditures on good Z

**14.** When Tom's income increases from $\$100$ to $\$120$, the quantity of golf balls he purchases increases from $10$ to $12$. What is Tom's income elasticity of demand, and what type of good is a golf ball for Tom?

(A) Income elasticity of demand is $1$, and golf balls are a normal good for Tom. ✓
(B) Income elasticity of demand is $-1$, and golf balls are an inferior good for Tom.
(C) Income elasticity of demand is $10$, and golf balls are a normal good for Tom.
(D) Income elasticity of demand is $-10$, and golf balls are an inferior good for Tom.
(E) Income elasticity of demand is $20$, and golf balls are a normal good for Tom.

**Answer A**

Correct. Income elasticity is calculated as the percentage change in quantity demanded/percentage change in income. Tom's quantity demanded increased by $20\%[= (\frac{12-10}{10} \times 100)]$ and Tom's income increased by $20\%[= (\frac{120-100}{100} \times 100)]$. Thus, Tom's income elasticity of demand is $1[= \frac{20\%}{20\%}]$. A positive income elasticity of demand indicates that golf balls are a normal good.

**15.** Assume that the price elasticity of supply for good Y is 0.5. If the price of good Y decreases by 30 percent, the quantity supplied of good Y will

(A) decrease by 60 percent
(B) decrease by 30 percent
(C) decrease by 15 percent ✓
(D) increase by 0.5 percent
(E) increase by 0.15 percent

**Answer C**

Correct. The price elasticity of supply measures the responsiveness of quantity supplied to a change in price and is calculated as the ratio of the percentage change in quantity supplied to the percentage change in price. Therefore, the percentage change in quantity supplied can be obtained by multiplying the price elasticity of supply (0.5) by the percentage change in the price (30 percent), which equals 15 percent.

## 2.4-2.5 other elasticities

**16.** Assume a 10 percent increase in price increased the market quantity supplied by 20 percent. Which of the following is true?

(A) The value of the price elasticity of supply is 2. ✓

(B) The value of the price elasticity of supply is 0.5.

(C) Supply is price inelastic.

(D) Demand is price elastic.

(E) This price-quantity combination violates the law of supply.

**Answer A**

Correct. The price elasticity of supply measures the responsiveness of quantity supplied to a change in price and is calculated as the ratio of the percentage change in quantity supplied to the percentage change in price. The value of the price elasticity of supply is 2, which is calculated as 20 percent divided by 10 percent.

**17.** If the value of the price elasticity of supply is 3, which of the following is true?

(A) Supply is inelastic.

(B) A percentage increase in price will lead to a relatively smaller percentage increase in quantity supplied.

(C) The supply curve is downward sloping with respect to the price of output.

(D) A 10 percent decrease in price will decrease the quantity supplied by 30 percent. ✓

(E) A 3 percent increase in price will decrease the quantity supplied by 10 percent.

**Answer D**

Correct. The price elasticity of supply measures the responsiveness of quantity supplied to a change in price and is calculated as the ratio of the percentage change in quantity supplied to the percentage change in price. When the magnitude of the price elasticity of supply is 3, a 10 percent decrease in price will lead to a 30 percent decrease in quantity supplied.

**18.** The cross-price elasticity of demand for flashlights and Good H is −2.5. This indicates which of the following?

## 2.4-2.5 other elasticities

(A) Flashlights and Good H are inferior goods.
(B) Flashlights and Good H are normal goods.
(C) Flashlights and Good H are substitutes in consumption.
(D) Flashlights and Good H are complements in consumption. ✓
(E) Flashlights and Good H both have elastic demand.

**Answer D**

Correct. Cross-price elasticity of demand is used to determine whether goods are substitutes, complements, or unrelated in consumption. When the cross-price elasticity coefficient is negative, the two goods are complements in consumption.

**19.** Following a decrease in the supply of oranges, the price of orange juice increased by 20 percent, which resulted in a 10 percent increase in the quantity of apple juice consumed. This implies that the cross elasticity of demand between orange juice and apple juice is

(A) 0
(B) 0.5 ✓
(C) 1
(D) 2
(E) indeterminate

**20.** Cross-price elasticity of demand for honey with respect to yogurt is $-4$. At the current equilibrium, consumers are buying 100 containers of yogurt per week and 20 containers of honey per week. If the price of yogurt increases by 10 percent per container, what will be the new quantity demanded of honey?

(A) 4 containers
(B) 8 containers
(C) 10 containers
(D) 12 containers ✓
(E) 40 containers

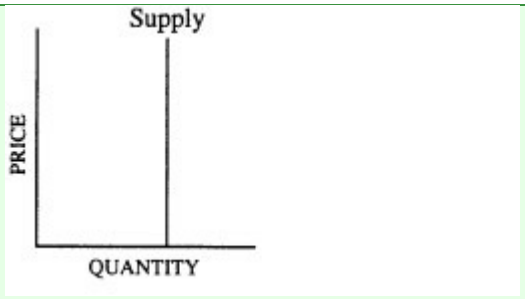
**Answer D**

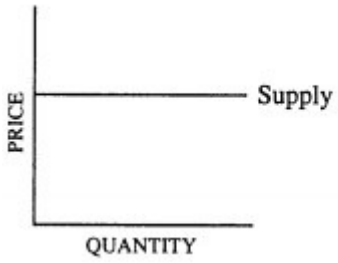
Correct. A negative cross-price elasticity of demand coefficient of $-4$ indicates that honey and yogurt are complements in consumption. Since the two goods are consumed together, an increase in the price of one good will cause a decrease in the demand, and therefore the quantity demanded, for the other good.

## 2.4-2.5 other elasticities

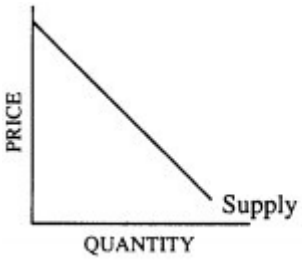
In this case, cross-price elasticity of demand for honey with respect to yogurt is equal to the % change in quantity demanded of honey / % change in the price of yogurt. Thus, a 10 percent increase in the price of yogurt resulted in a 40 percent decrease in the quantity demanded of honey, i.e. quantity demanded of honey will decrease by 8 containers, and therefore the new quantity demanded of honey will be 12 containers.

**21.** If a one-of-a-kind Etruscan vase is offered for sale at an auction, which, if any, of the following correctly shows the supply curve for the vase?

(A) 


(B) 


(C) 


(D) 


(E) It is impossible to determine the shape of the supply curve from the given information.

## 2.4-2.5 other elasticities

**22.** If the income elasticity of demand for good X is negative and the cross-price elasticity of demand between good X and good Y is negative, which of the following must be true of good X?

(A) X is a normal good and is a substitute for Y.

(B) X is a normal good and is a complement to Y.

(C) X is an inferior good and is a substitute for Y.

(D) X is an inferior good and is a complement to Y. ✓

(E) X is a normal good and Y is an inferior good.

**23.** Assume the income elasticity of demand for good Z equals $-5.0$. Which of the following is true?

(A) Good Z is a normal good.

(B) Good Z must have an inelastic demand.

(C) An increase in income will lead to a decrease in demand. ✓

(D) An increase in income will lead to an increase in demand.

(E) The income effect of a price increase will be a decrease in quantity demanded at every price.

**Answer C**

Correct. Income elasticity of demand measures the responsiveness of demand to changes in consumers' income. When the coefficient of the income elasticity of demand is negative, this indicates that Good Z is an inferior good—that is, the demand for Good Z decreases as consumers' income increases.

**24.** Which of the following statements relating to income elasticity is true?

(A) A positive value for the income elasticity coefficient indicates an inferior good.

(B) If good X and good Y have negative income elasticities, then both goods are substitutes.

(C) With an income elasticity coefficient of 0.6, the demand is inelastic and the good is an inferior good.

(D) With an income elasticity coefficient of 5, a 10 percent increase in income will lead to a 50 percent increase in the quantity demanded of the good. ✓

(E) With an income elasticity coefficient of $-1.2$, a 10 percent increase in income will lead to a 12 percent decrease in the price of the good.

**Answer D**

Correct. Income elasticity of demand measures the responsiveness of demand to changes in consumers' income and is calculated as the ratio of the percentage change in quantity demanded to the percentage

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change in income. When the magnitude of the income elasticity is 5, a 10 percent increase in income will lead to a 50 percent increase in quantity demanded of the good.

**25.** A 10 percent increase in Sandra's income causes Sandra's consumption of milk to decrease from 10 cartons to 6 cartons. Sandra's income elasticity of demand for milk is

(A) greater than zero and therefore milk is a necessity
(B) greater than zero and therefore milk is a normal good
(C) less than zero and therefore milk is a normal good
(D) less than zero and therefore milk is an inferior good ✓
(E) greater than one and therefore milk is a necessity

**Answer D**

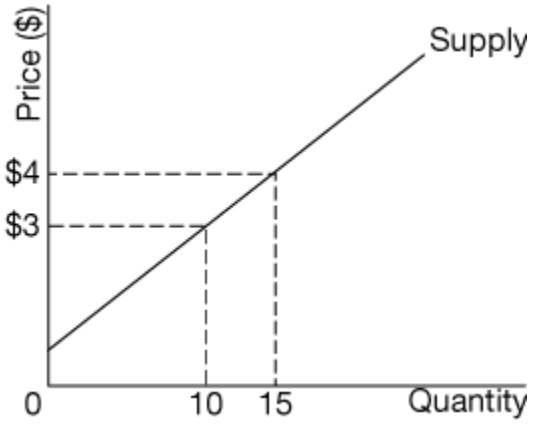
Correct. Sandra's income elasticity of demand for milk is $\frac{\left[\left(\frac{10-6}{10}\right)\times 100\right]}{10\%} = \frac{-40\%}{10\%} = -4$ which is less than zero. A negative income elasticity of demand indicates that milk is an inferior good.

**26.** The quantity of peanuts supplied increased from 40 tons per week to 60 tons per week when the price of peanuts increased from \$4 per ton to \$5 per ton. The price elasticity of supply for peanuts over this price range is

(A) elastic ✓
(B) inelastic
(C) unit elastic
(D) perfectly elastic
(E) perfectly inelastic

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**27.**




What is the price elasticity of supply between $\$3$ and $\$4$ in the graph above?

(A) It is greater than $1$, and the supply is relatively inelastic.

(B) It is greater than $1$, and the supply is relatively elastic. ✓

(C) It is equal to $1$, and the supply is unit elastic.

(D) It is less than $1$, and the supply is relatively elastic.

(E) It is less than $1$, and the supply is relatively inelastic.

**Answer B**

Correct. The formula to calculate price elasticity of supply is the percentage change in quantity divided by the percentage change in price, which in this case is $= \frac{\frac{15-10}{10} \times 100}{\frac{\$4-\$3}{\$3} \times 100} = \frac{50\%}{33\%} = 1.52$. The value is greater than one, so supply is relatively elastic.

**28.** Suppose the price elasticity of supply for gasoline in the short run is estimated to be 0.4. Due to an unexpected surge in the demand for gasoline, the price of gasoline increases by 20 percent. As a result, the quantity supplied of gasoline will

(A) increase by 50 percent

(B) increase by 20 percent

(C) increase by 8 percent ✓

(D) decrease by 0.2 percent

(E) be impossible to determine from the given information

## 2.4-2.5 other elasticities

**29.** **The supply schedule gives two points on a market supply curve.**

| Price | Quantity Supplied |
|---|---|
| $10 | 50 |
| $20 | 60 |

Which of the following is true about the supply curve between the given points?

(A) The supply curve is inelastic, because the percentage change in the price is greater than the percentage change in the quantity supplied. ✓

(B) The supply curve is unit elastic, because the change in price is $10 and the change in quantity supplied is 10 units.

(C) The supply curve is perfectly elastic, because the change in price is $10 and the change in quantity supplied is 10 units.

(D) The supply curve is elastic, because the price elasticity of supply is equal to 10.

(E) The supply curve is unit elastic, because the price elasticity of supply is equal to 1.

**Answer A**

Correct. Recognizing percentage changes is what matters for elasticity. The supply is inelastic. It can be determined by comparing the relative changes in price and quantity. Price increases by 100 percent, and the quantity supplied increases by 20 percent. Thus, 20 divided by 100 is equal to 0.2, which is inelastic.

**30.** Which of the following would cause the supply of good X to become more elastic?

(A) Greater availability of substitutes for good X

(B) Increased prices of inputs required to produce good X

(C) The ability to easily reallocate inputs to production of good X ✓

(D) A short time frame for making production decisions

(E) More elastic demand for good X

**Answer C**

Correct. Elasticity is a measure of responsiveness of quantity supplied to changes in price. The ability to reallocate resources easily allows producers to respond to changes in price quickly. One of the determinants of the elasticity of supply is the time period it takes to respond to changes in market

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conditions. The ability to easily reallocate inputs lessens the time period to adjust and respond. Thus, supply becomes more elastic.

**31.** If the value of the price elasticity of supply for a product is 0.5, which of the following statements is correct?

(A) A 10 percent increase in the price of the product will decrease the quantity supplied by 5 percent.
(B) A 10 percent increase in the price of the product will decrease the quantity supplied by 20 percent.
(C) A 10 percent increase in the price of the product will increase the quantity supplied by 20 percent.
(D) A 20 percent decrease in the price of the product will decrease the quantity supplied by 10 percent. ✓
(E) A 20 percent decrease in the price of the product will decrease the quantity supplied by 40 percent.

**Answer D**

Correct. The formula for the price elasticity of supply is given by percentage change in quantity supplied ⁄percentage change in price. Therefore, when the price elasticity of supply is 0.5, a 20 percent decrease in price results in a 10 percent decrease in quantity supplied.

**32.** A market supply curve is more elastic in the long run than in the short run because

(A) producers' marginal costs are increasing
(B) the quantity demanded increases at a steady rate
(C) there are no fixed inputs in the long run ✓
(D) of constant returns to scale in production
(E) of diseconomies of scale in production

**Answer C**

Correct. Since there are no fixed inputs in the long run, firms can more easily expand or contract their operations in the long run. In the long run, all inputs, including land and physical capital, are variable.